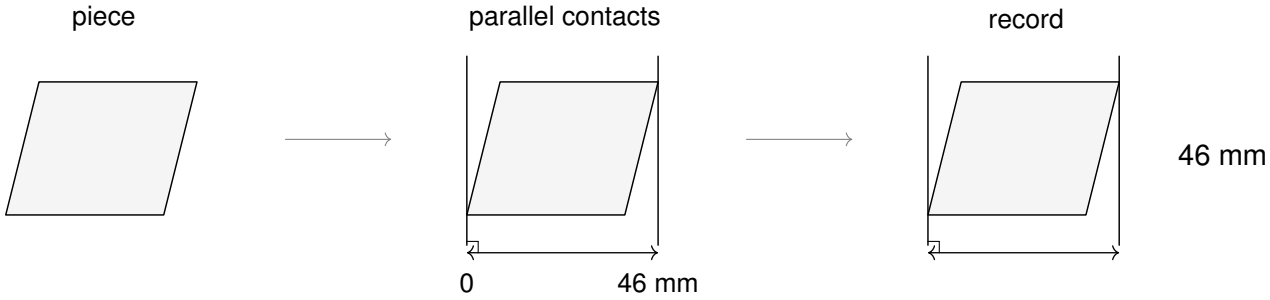
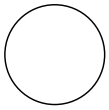
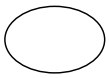
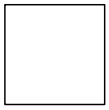
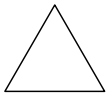
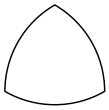
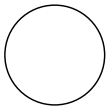
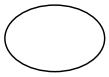
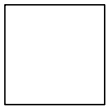
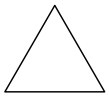
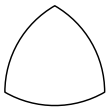
To measure width, keep the two straight edges parallel, enclosing the whole piece and just touching it. Turn and slide the piece freely. Measure the perpendicular gap, to the nearest millimeter. This gap is the width in that position. Drawings are smaller than the real pieces.



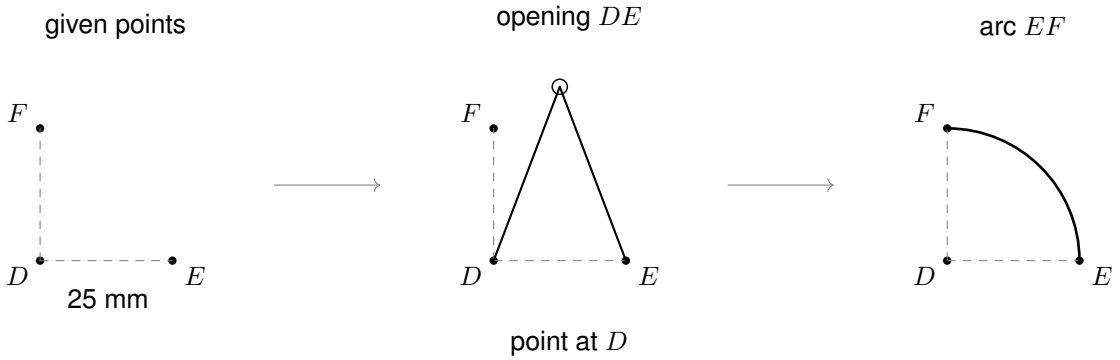
Problem 1: Use the five pieces. Find positions that make each gap as small and as large as you can. Keep a sketch and the gap for each.

	smallest gap	largest gap
 disk		
 oval		
 square		
 straight triangle		
 curved triangle		

Problem 2: Close the edges on the disk, then keep them fixed while the piece turns and slides. Which pieces can turn all the way around inside them? Which can keep touching both edges throughout the turn? Keep a sketch of a deciding position.

	turn inside?	touch both throughout?	deciding position
 disk			
 oval			
 square			
 straight triangle			
 curved triangle			

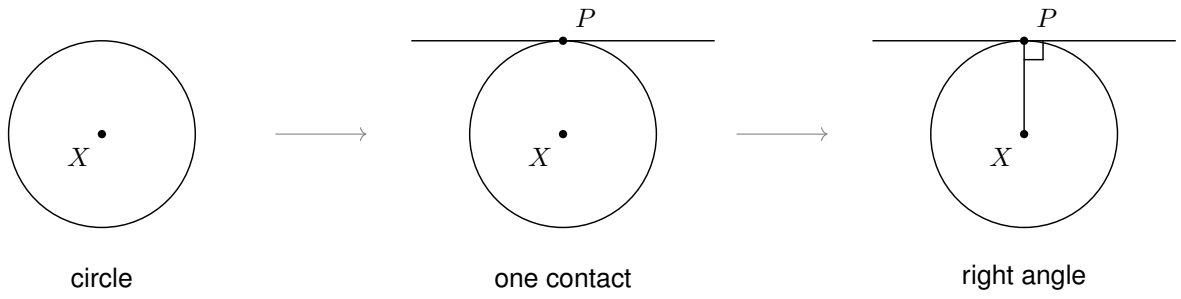
A compass draws an arc with one center and one radius.



Problem 3: Use the 60 mm and 40 mm equilateral templates to make two curved triangles. Each short arc joins two corners and has the third corner as its center. Predict the width of each piece, then test it.

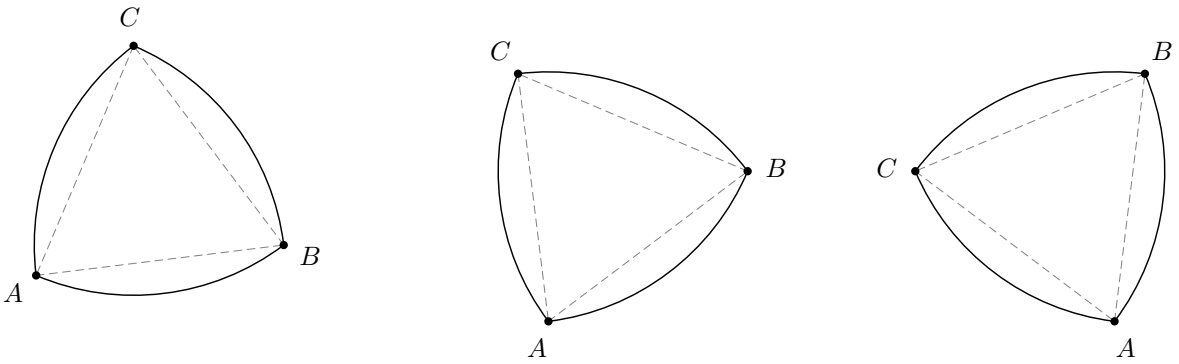
60 mm triangle	40 mm triangle
predicted width:	predicted width:
measured gaps:	measured gaps:

A line that touches a circle at just one point meets the radius there at a right angle.

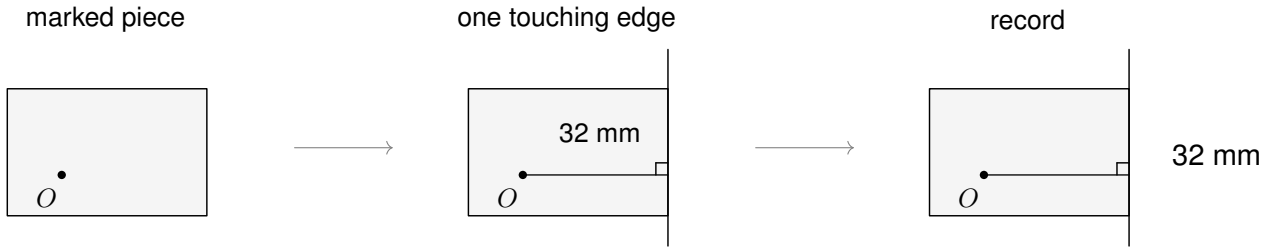


circle one contact right angle

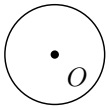
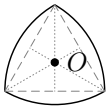
Problem 4: Explain why the exact curved triangle has the same width in every direction. Its arcs have radius 60 mm, with centers at A, B and C. Your explanation should also work for every position between the drawings.



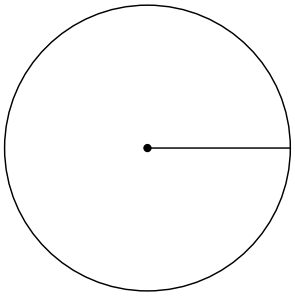
The mark O is a point on a piece. Measure to one straight edge that encloses the whole piece and touches it.



Problem 5: Use the marked disk and curved triangle. Find the smallest and largest perpendicular distances from O to one touching edge. Can a piece have constant width while those distances change?

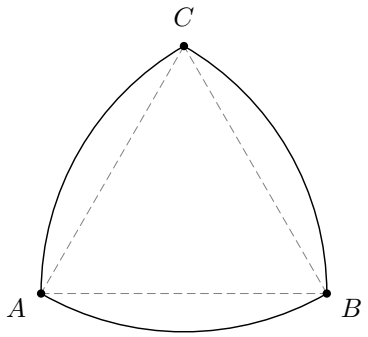
	smallest O-to-edge distance	largest O-to-edge distance
 marked disk		
 marked curved triangle		

Problem 6: Compare the boundary lengths of the 60 mm-width disk and curved triangle. Explain how you know.



radius 30 mm

disk



arcs: radius 60 mm

curved triangle